

CyberHost Is the First One-Stage Audio-Driven Talking Body With Zero-Shot Generalization

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TAKEAWAYS

- One-stage diffusion for audio-driven talking body; extends to video-/multimodal-driven.
- Region Attention Module (RAM) + Human-Prior-Guided Conditions improve hands/face and motion stability.
- Outperforms prior work in FID/FVD, identity, and sync; zero-shot on open-set images.

* Equal contribution | † Project lead | ‡ Work done during internship

<https://cyberhost.github.io/>

Audio Alone Underfits Local Details and Induces Uncertain Motion

DETAILS UNDERFITTING

- Audio lacks explicit local control; hands/face are small but identity-dense.
- Degraded hand integrity and facial details in half-body settings.

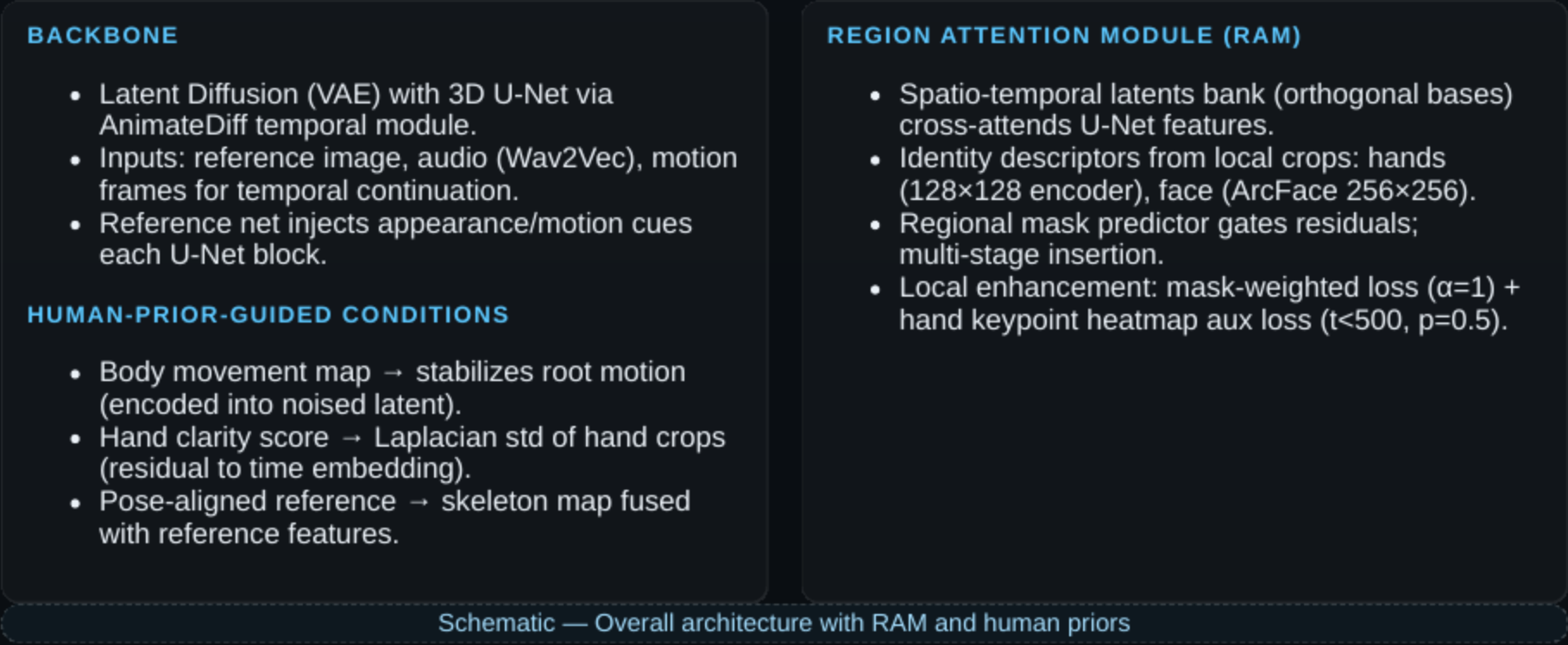
MOTION UNCERTAINTY

- Weak audio–limb correlation → one-to-many motion; unstable global/body movements.
- Probabilistic hand blur from data further harms learning.

Problem framing

Targets: details + stability

One-Stage Diffusion Matches Lip Sync While Fixing Hand and Face Detail



Curriculum Training Enables Stable 200h Scaling

- Stage-1: two frames (reference → target); learn reference net, pose encoder, base U-Net.
- Stage-2: end-to-end audio → video; optimize temporal/audio layers and RAMs.
- Clips: 12 frames; motion frames: 4; lr=1e-5; 640×384 then dynamic area≈640×384.
- Data: ~200h half-body speech, 10k+ identities; test: 269 segments from 119 IDs.

CyberHost Leads on FID/FVD While Preserving Identity and Lip Sync

TINY LEGEND

- FID/FVD: image/video quality (↓ better)
- CSIM: facial identity similarity (↑)
- SyncC/SyncD: audio-lip sync (↑/↓)
- HKC/HKV: hand clarity/motion richness (↑/↑)

Mini radar/bar — FID↓ FVD↓ CSIM↑ SyncC↑ HKC↑

LARGE QUALITATIVE ANCHOR

Side-by-side: Ours vs Two-Stage (hand close-up, face close-up)

Sharper fingers, stable gaze, consistent identity across frames.

CyberHost Produces Sharper Hands and More Consistent Faces Than Baselines

A2V-B: Ours vs DiffTED (hands)

Fewer finger artifacts; higher HKC.

A2V-B: Ours vs DiffGesture+MimicMotion

Cleaner palms; stable wrist pose.

Vs Vlogger demo (face)

Higher CSIM; preserved identity.

Vs Vlogger demo (hands)

Natural gesture diversity (↑HKV).

Close-up: lips sync

Improved SyncC; reduced SyncD.

Close-up: hand motion blur

Hand clarity prior yields crisp frames.

Open-set portrait → body

Zero-shot gesture integrity.

Reference pose challenge

Pose-aligned ref improves hands.

Generalizes to Video-Driven and Open-Set Images Without Fine-Tuning

VIDEO-DRIVEN REENACTMENT (V2V-B)		OPEN-SET & MULTIMODAL	
Ours vs AnimateAnyone Better hand structure; fewer artifacts.	Ours vs MimicMotion Lower AKD; cleaner faces.	Internet/AIGC subject Consistent ID, complex gestures.	Audio + hand keypoints Explicit hand control; synced lips.
Ours vs DisCo Higher temporal coherence (↓FVD).	Skeleton → Video sample Accurate body transfer, intact hands.	Challenging lighting Stable gaze; low flicker.	Fast gestures Reduced motion blur; crisp fingers.

Failure Cases Reduced by RAM and Human Priors

COMMON FAILURES (BASELINES)		CYBERHOST IMPROVEMENTS	
Baseline: finger melt Loss of topology under fast motion.	Baseline: off-sync lips Drifts during long utterances.	RAM hands Crisp fingers; preserved joints.	Sync-stable speech Higher SyncC; smoother visemes.
Baseline: ID drift Facial texture inconsistency.	Baseline: root wobble Unstable torso translation.	Pose-aligned ref Robust initial hand pose.	Movement map Stable root; natural sway.

Design impact	Hands/face + motion stability

RAM Improves Hand Clarity; Human Priors Stabilize Motion

REGION ATTENTION

- Latents bank → ↑HKC/HKV; temporal latents vital for motion diversity.
- Identity descriptors → ↑CSIM and facial consistency.
- Face RAM → better CSIM/Sync; Hand RAM → fewer artifacts (↓FID/↓FVD).

HUMAN-PRIOR CONDITIONS

- Without movement map: unstable global motion (↓SSIM/PSNR, ↑FVD/FID).
- Without hand clarity: erratic/blurry hands (↓HKC; inflated HKV).
- Without pose-aligned ref: worse hand quality and higher FID/FVD.

Tables 1–3 (paper) report exact values; our variants confirm each component's effect.

CyberHost Delivers Stable, Detailed, Identity-Consistent Talking Bodies

AUDIO-DRIVEN

Result sample A

Natural motion; synced lips.

Result sample B

Sharp hands; stable gaze.

Result sample C

Consistent identity across time.

ZERO-SHOT & OPEN-SET

Unseen subject

Preserved appearance.

AIGC image

Complex gestures handled.

Challenging pose

Hands remain intact.